



BASHA

Muhammad Hawatif Basharuddin Rabbani Tahammal

Software Engineer

[bashatahamal.github.io](https://github.com/bashatahamal)

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Summary — A highly skilled Software Engineer with a strong foundation in machine learning, software development, and DevOps. With nearly 4 years of professional experience, I have successfully led and developed AI-driven applications, including LLM-based conversational assistants, OCR automation, computer vision solutions, and recommendation systems. My expertise spans AI applications, Dev/MLOps, cloud optimization, and AI-driven automation, with a proven track record of enhancing efficiency and reducing operational costs. Passionate about integrating AI solutions into real-world applications, I have contributed to multiple industries, including aquaculture, cybersecurity, e-commerce, and agribusiness. I excel in cross-functional teams, leveraging technical expertise, leadership, and innovation to drive impactful and scalable AI solutions.

Experience

eFishery

July 2023 – Present

Mid AI Research Engineer

Bandung, ID

eFishery is Indonesia's first aquaculture technology unicorn and one of the largest companies in the sector globally. It provides innovative solutions for fish and shrimp farming, aimed at enhancing efficiency, productivity, and sustainability in aquaculture.

AI Research Engineer

– LLM Applications

- Initiated and led AI research projects on Large Language Models (LLM), starting with functionality in Retrieval-Augmented Generation (RAG) and real-time database integration. The project later became known as [Mas Ahya](#).
- Created, engineered, and led the breakthrough development of Mas Ahya (a generative-AI-powered conversational assistant for aquaculture). As the principal engineer of Mas Ahya, I maintained and improved the core system, releasing it for internal farmer use. It was presented at [Microsoft AI Day](#) by Microsoft CEO [Satya Nadella](#) and featured in various [news outlets](#) and [social media platforms](#).
- Developed internal LLM application, enabling the internal shrimp team to receive better recommendations on which ponds are ready for harvest via conversational interactions.
- Initiated and led the introduction of GPT automated testing to enhance engineering processes, specifically in QA testing and evaluating the performance of LLM applications.
- Inspired the development of an OCR LLM-based solution that reduced and saved engineering costs.
- Planned LLM Apps MultiAgent systems roadmap and conducted proof of concept (PoC) with capabilities integration with WhatsApp chat/non-text media, including voice (text to speech, speech to text), image, and multi-step data confirmation.

– MLOps

- Achieved up to 80% cost reduction in LLM services operations through performance optimization and resource efficiency improvements.
- Led cloud cost analysis, implementing strategies that resulted in up to 60% monthly cost savings through usage optimization.
- Ensured smooth deployment and operational efficiency of AI services.
- Conducted AI service assessments, identifying and addressing areas for cost reduction and performance enhancement.
- Contributed to designing end-to-end Edge Devices and AI roadmap for seamless integration and deployment.

AI Squad Leader

- Squad & Tech Implementation Leadership
 - Led ML/AI development driving cross-functional collaboration with engineers, PMs, and stakeholders.
 - Managed sprint planning, reviews, and retrospectives to ensure efficient Agile execution.
 - Oversaw the implementation and maintenance of ML/AI services across AWS, GCP, Azure.
 - Enforced code review standards and integrated automated quality control tools for reliability and scalability.
 - Ensured seamless ML service deployment across cloud environments.
- Technical Council Contribution
 - Regular speaker to the Tech Council, a bi-weekly multi-division forum involving VP, engineers, and PMs.
 - Critiqued and reviewed technical proposals and implementation strategies to ensure optimal execution.
- Supervised AI Projects
 - Optical Character Recognition (OCR) Solutions
 - Automated sensitive data masking for bank statements (BNI, BRI, BCA, CIMB), processing up to 500-page documents with RabbitMQ for efficient workload distribution and manual error reduction.
 - OCR enhancement with GPT-4.0, cutting processing time, manual data collection, and engineering costs.

- Document detection and classification for eKYC, utilizing YOLOv8 to optimize ID verification while reducing unnecessary OCR processing costs through manual filtering.
- Custom OCR system with Azure Document Intelligence, replacing third-party services and cutting costs.
- **Tools Used:** Python, Langchain, Langgraph, Langfuse, FastAPI, AWS, GCP, Azure Platform, Nomad, Kubernetes, GPT, LLM, Ollama, Huggingface, Alembic, Pandas, Polar, PyTorch, PostgreSQL, MongoDB etc.

HDCE Inc.

October 2024 – December 2024

Freelance AI Engineer

Canada, CA

A cybersecurity consulting firm headquartered in Canada specializes in identifying potential threats and vulnerabilities. It provide actionable recommendations and solutions to mitigate any security risks.

AI & Machine Learning Training

- Mentored and guided tutors on local LLM model training, fine-tuning, and implementation, providing hands-on support for real-world applications.

Proof of Concept Development

- Designed and implemented a phone call voice-based authentication system, integrating voice recognition, biometric security, speech-to-text, text-to-speech and Twilio's API for secure and scalable deployment.
- **Tools Used:** Python, PyTorch, CUDA, Scikit-learn, Huggingface, Twilio, Flask

Great Giant Foods

July 2024 – September 2024

Freelance AI Engineer

Yogyakarta, ID

Great Giant Foods (GGF) is a leading agribusiness company in Indonesia, specializing in the production and export of high-quality fresh and processed fruits, juices, meats, and dairy products.

OCR Based Solution

- Developed OCR-based desktop applications to enhance the efficiency of banana harvest form record-keeping using GPT-4o, ensuring easy maintenance by optimizing prompt engineering assets.
- Developed API-based OCR Services leveraging GPT-4o for seamless integration and communication with other services.

LLM Based Application

- Built and Deployed End-to-End RAG-based (Retrieval Augmented Generation) applications using GPT-4o with a Streamlit web application to provide internal educational resources on banana cultivation, backed by grounded data knowledge.
- **Tools Used:** Python, FastAPI, Langchain, OpenAI, Streamlit, Pandas, PyQt5, Docker

Yuna & Co.

April 2021 – June 2023

Machine Learning & Software Engineer, DevOps

Jakarta, ID

A women's fashion and lifestyle e-commerce platform & personal styling based on Artificial Intelligence.

Machine Learning related responsibility

- Developed a user clustering service for enhanced data analysis.
- Created a product recommendation service based on similar products, with a caching strategy for efficient output.
- Implemented image understanding functionalities, such as object detection for duplicate image search, style card generator functionality (outfit reference images containing recommended products), and color picker based on similar RGB values.
- Developed a service for product searches based on similar characteristics using images and/or tagging. This was utilized for functionalities such as user wardrobe (personal collection and recommended outfits), style card generator (admin tool for faster generation of style cards), and similar product suggestions for the main website shop page.
- Implemented collaborative filtering to recommend products based on user feedback and constructed generic user profile characteristics.
- **Tools Used:** Python, Scikit-learn, OpenCV, Google Vision, Numpy, Pandas and other related libraries/frameworks.

Software Engineer related responsibility

- Built and maintained machine learning, image, and data API services using Python FastAPI.
- Developed and managed task scheduler services using Celery, Django, MongoDB, and RabbitMQ.
- Responsible for building and maintaining Python API services and contributing to NodeJS API maintenance.
- Designed, created, and maintained asset management services with image manipulation capabilities (padding, cropping, resizing, color changing) and cached derivative images for faster load times.
- Designed, maintained and implemented a message broker for communication between services using RabbitMQ.
- Contributed to debugging and migrating the NodeJS API application to a new Typescript framework.
- Added listeners and notifiers to the PostgreSQL database for data validation.
- Managed data migration, cleansing and provided data summaries for further analysis.
- **Tools Used:** Python, OpenCV, Flask, FastAPI, Django, Celery, Pandas, RabbitMQ, Postgres, MongoDB, NodeJS, TypeORM, Typescript

DevOps related responsibility

- Managed and handled all aspects of development operations, including deployment, server management, application containerization, automation and etc.
- Initiated and implemented the use of Kubernetes for application deployment along with CI/CD pipeline deployment using Jenkins, Ansible, and related frameworks.
- Dockerized and containerized all Yuna applications for microservices deployment into GCP and AWS.
- Created, designed, tested, and maintained infrastructure for all services and microservices-based applications using Kubernetes.
- Handled network traffic using Ingress, Nginx, load balancers, and Cloudflare.
- Built automation tasks such as Selenium testing, Telegram Bot - Jenkins integration (sending Jenkins commands via button in Telegram chat group), updating image product sets on Google Vision, automatic database backups, and WhatsApp server automation.
- **Tools Used:** Google Cloud Platform (GCP), Amazon Web Services (AWS), Kubernetes, Ansible, Terraform, Kubernetes Operations (KOps), Jenkins, Cloudflare, Telegram bot UI, Bitbucket webhook, Selenium, PyWinAuto, Bash Script

Freelance Project

September 2020 – February 2021

Professional Assignments (ML Researcher)

Yogyakarta, ID

Traffic monitoring system for counting vehicles and detecting traffic violation based on video feed

Intelligence Traffic Monitoring System

- Led the project, providing strategic solutions and innovative ideas.
- Prepared, trained, and fine-tuned the object recognition model using a custom low-resolution dataset.
- Developed a web-based system interface.
- Implemented algorithms for traffic violation detection and vehicle counting through video processing (object recognition & object tracking).
- **Tools Used:** Python, Tensorflow, PyTorch, OpenCV, Flask, Javascript, HTML/CSS

Freelance Project

July 2020

Professional Assignments (ML Researcher)

Malaysia, MY

Detect and recognize road damage on images pixel by pixel and put mark/bounding box on it.

Road Damage Detection and Recognition

- Implemented methods to identify and classify road damage types using Yolov4 and Yolov5 Convolutional Neural Network architectures.
- Trained and fine-tuned architecture models with the prepared dataset.
- Compared the latest methods for each required output type using LinkNet, UNet, and Feature Pyramid Network (FPN) with varied Convolutional Neural Network backbones.
- Compiled final reports for testing and model evaluation.
- **Tools Used:** Python, Tensorflow, PyTorch, OpenCV, Google Colaboratory

Grant-Funded Project

August 2019 - December 2019

Innovation and Creativity Project in Indonesian Higher Education

Yogyakarta, ID

Road block-like tools on pedestrian paths which function to prevent other than pedestrians and cyclists from entering the area using real time video processing.

Pedestrian Detector for Sidewalk Safety

- Project leader and programmer.
- Tuned pre-trained Convolutional Neural Network models for object detection.
- Configured Raspberry Pi and Intel Movidius Neural Compute Stick as the main controller.
- Developed complete code for both hardware and software components.
- Designed algorithms and electronic circuits.
- **Tools Used:** Python, Tensorflow, OpenCV, Raspberry Pi, Intel Movidius NCSDK

Education

Gadjah Mada University (UGM)

August 2016 - November 2020

Bachelor of Electronics and Instrumentation

GPA 3.61 out of 4.00

Minors: Computer Vision; Machine Learning

Undergraduate thesis project

- **Topic:** Quran Tajweed Rules Recognition
- Conducted research to develop methods for recognizing Tajweed rules using image-based input (document processing).
- The end-to-end project involved designing the method and its evaluation scheme, building a dataset of Arabic characters, implementing image processing techniques consisting of detecting the occurrence of reading law characters using template matching, developing a segmentation algorithm for Arabic words based on pixel values, and utilizing Convolutional Neural Networks for character recognition. Additionally, a web-based user interface was developed.
- **Tools Used:** Python, Tensorflow, OpenCV, Flask, PySimpleGUI, Numpy, Pandas, Javascript, HTML/CS

Skills

AI/ML	LLMs (RAG, Langchain, Langgraph), Computer Vision (OCR, Object Detection, Image Processing), NLP, Speech Recognition, Recommendation Systems	Automation	Celery, Bash, Chatbot Integration (WhatsApp, Telegram), OCR Services, LLM evaluation
Backend	Python (FastAPI, Flask, Django), TypeScript (Node.js), API Development, Microservices, Event-Driven Systems	Common Libs	PyTorch, TensorFlow, Huggingface, OpenAI, Pandas, NumPy, OpenCV, Langfuse, Ollama, Streamlit etc.
Databases	PostgreSQL, MongoDB, VectorDB	Cloud Platform	Azure Platform, Google Cloud Platform, Amazon Web Services
Messaging	RabbitMQ, Celery	Soft Skills	Fast learner, Leadership, Agile, Collaboration, Technical principal, Strong problem-solving skills
DevOps	AWS, GCP, Azure, Docker, Kubernetes, Nomad, Terraform, Ansible, Jenkins, Bitbucket Pipelines, Git	Languages	English, Bahasa Indonesia